

# CorpusMind v1.2.0 — User Guide

The two-page quick start: corpora, the analysis battery, the new Learner Research suite, Vector KWIC, and the companion apps.

- NEW · Learner Research (CAF · CIA · Errors)
- NEW · Vector KWIC (bge-m3)
- NEW · Hugging Face explorer
- EN / عربي

waleedmandour.org · September 2026 · CorpusMind (main app) v1.2.0 · Companion: CorpusMind Lens

## 1 · GETTING STARTED

### From raw texts to analysis in three steps

CorpusMind is a **local-first** research environment: your data never leaves your machine, and the grounded assistant answers only with evidence from your own corpora. Open **Your Corpus** and work top to bottom:

- 1 Create a project & corpus

Optional learner facets: the learners' **L1** and a **CEFR level (A1–C2)** at creation — they power the new group-by views.
- 2 Upload texts

Drag in TXT, DOCX, PDF, HTML or XML — the engine builds a versioned annotation layer (tokens, lemmas, POS, morphology).
- 3 Pick a tool

Concordance, Frequency, Collocation, Keyness, N-grams, Dispersion and more — every result carries its full parameters.

Tip: press **Ctrl/Cmd + K** for the command palette. The floating AI assistant (bottom-right) is context-aware on every tool screen — in v1.2.0 its panel is wider and wraps model output properly.

## 2 · THE WORKSPACE

### One sidebar, six working groups

|                  |                                                                                                                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corpora          | Your Corpus (create, upload, clean, subcorpora) and Reference Corpus for comparisons. Vision corpora now live in the Lens companion — the parent-app Vision tab was retired in v1.2.0, and stale links redirect automatically. |
| Analysis Tools   | Concordance, Frequency, Collocation, Keyness, Dispersion, Documents, Readability, Groups, N-grams, POS, Grammar, Discourse, Vocabulary, Sentiment, Metaphor — plus <b>Vector KWIC</b> (§3).                                    |
| Learner Research | <div>NEW</div> أبحاث لغة المتعلم — CAF Report, CIA Compare and Error Patterns for learner-corpus work (§4).                                                                                                                    |
| Arabic           | Morphology, roots, clitics, Buckwalter transliteration, dialect and register identification — Arabic is first-class throughout the suite (§5).                                                                                 |
| AI & System      | The grounded AI Assistant; Settings with the new Hugging Face model explorer (§6); User Guide; About.                                                                                                                          |

## 3 · NEW ANALYSIS TOOL

### Vector KWIC — concordancing with embeddings NEW

Following Anthony (2025), **Mode A** takes a node word plus a research query: every KWIC line containing the node is re-ranked by cosine similarity between its context window and your query, so the lines that *mean* what you are looking for surface first. **Mode B** leaves the node empty and just asks a question — the corpus is scanned sentence-by-sentence (capped at 6,000, always reported) for the top-k most similar sentences.

- One-click setup — and an honesty rule

Lines are embedded with a local model — recommended: `ollama pull bge-m3` (multilingual, strong Arabic), also in Settings → Models → Embedding; a missing model triggers a one-click pull hint. The **Similarity** column is **raw cosine similarity** — closer means more relevant to your query, never a confidence score. Vectors are cached per corpus and model; exports include similarity.

## Learner Research — CAF, CIA and error candidates NEW

The **Learner Research** group (أبحاث لغة المتعلم) brings learner-corpus methodology into the app — exportable, with formulas and citations shown in the interface.

### CAF Report

Complexity–Accuracy–Fluency per corpus or grouped by **L1** / **CEFR**: TTR, MATTR, MTLD, HD-D, Guiraud; EN clause indices from the dependency parser; Arabic word-length and root-richness proxies; a labelled error-free-sentence heuristic.

### CIA Compare

Contrastive Interlanguage Analysis (Granger 1998): learner corpus vs reference (engine corpus or bundled list), or group vs group — keyness plus CAF deltas in one exportable report.

### Error Patterns

Rule-based error **candidates**, never confirmed errors: EN articles, prepositions, agreement, spelling; AR hamza variants, ه/ة and ي/ى confusions. Each links to its exact line — you verify before counting.

### AI-vs-learner comparator

Paste an AI text on the same prompt: the identical CAF battery runs on it — a side-by-side delta table against your learner corpus (cf. the 2025 LLM-vs-student-writing studies).

## 5 · ARABIC-FIRST SEARCH

### Arabic normalization across the tools NEW

The Concordancer, Frequency, Collocation and Keyness tools now carry an **“Arabic normalization”** toggle: hamza spellings unify (ا → آ إ إ), teh marbuta (ه → ة), alef maksura (ي → ى), and diacritics are stripped *before* matching or counting — spelling variants collapse into one row without re-ingesting anything. Regex always runs on raw text (and the tool says so); the same flag exists inside Vector KWIC.

## 6 · MODELS & AI

### Settings: curated lists, Hugging Face, fit badges NEW

Settings → Models separates **text** and **embedding** models with filter chips (bge-m3 is the recommended embedder); vision models moved to Lens. The new **Hugging Face explorer** searches live GGUF models sorted by real downloads, shows actual per-quantisation file sizes, and labels each with a machine-aware fit badge — **GPU · CPU · Tight · Too-big** — from real RAM/VRAM (rule-of-thumb, never a guarantee). Pulling (hf.co/<user>/<repo>:<quant>) reuses the familiar pull-progress flow.

## 7 · THE CORPUSMIND FAMILY

### Companion apps: Lens & Voice NEW

CorpusMind is deliberately modular: the main app masters text; the companions extend it to images and audio — all three read the same engine and data directory, so projects, corpora and settings are shared.

#### CorpusMind Lens — vision

Vision-LM multimodal discourse analysis: image sets with OCR, Visual Grammar (Kress & van Leeuwen), image–text alignment, and a cross-modal AI that reads texts and images together. The parent-app Vision tab was removed in v1.2.0 — the workbench lives here now.

[waleedmandour.org/projects/CorpusMindLens](https://waleedmandour.org/projects/CorpusMindLens)

#### CorpusMind Voice — audio

Audio-to-corpus: transcribe recordings, build spoken-corpus data, and analyse it with the same battery of tools — closing the loop between written and spoken data in one research environment.

[waleedmandour.org/projects/CorpusMindVoice](https://waleedmandour.org/projects/CorpusMindVoice)

**References.** Anthony (2025) *Applied Corpus Linguistics* 5(3):100164 · Granger (1998) *Learner English on Computer* · McCarthy & Jarvis (2010) *Behav. Res. Methods* 42(2) · Housen & Kuiken (2009) *Applied Linguistics* 30(4) · Lu (2012) *MLJ* 96(2). · **CorpusMind** — AGPL-3.0 · local-first, no telemetry · docs/METHODOLOGY.md · [github.com/waleedmandour/CorpusMind](https://github.com/waleedmandour/CorpusMind)